

Homework 4 — Full Solutions (instructor / RA)

Numbers are from the provided data; last-decimal values are seed- and N_DRAWS-dependent, but signs, magnitudes, and patterns are stable.

1. Part 1 — pipeline output (expected numbers)

Demand (run_demand). Estimates recover the DGP: fare $\alpha \approx -0.012$, freq ≈ 0.39 , premium ≈ 0.49 , service ≈ 0.30 ; $\sigma_{\text{const}} \approx 0.30$, $\sigma_{\text{price}} \approx 0.006$; $\pi_{\text{inc} \rightarrow \text{const}} \approx 0.35$, $\pi_{\text{inc} \rightarrow \text{price}} \approx 0.004$. Mean own-fare elasticity ≈ -2.8 ; mean Lerner ≈ 0.39 ; all $mc > 0$. AA-BB symmetrized diversion on head-to-head routes $\bar{D}_{AB} \in [0.11, 0.27]$.

Costs (run_supply). $\log mc = \gamma'W + \omega$: fuel ≈ 0.21 , fees ≈ 0.09 , distance ≈ 0.29 , freq ≈ 0.09 , premium ≈ 0.17 , service ≈ 0.00 . Floored- mc share = 0%; $\text{sd}(\omega) \approx 0.09$.

Fixed-cost bounds (run_fixedcosts). Median upper bound (offered) ≈ 15 –16; median lower bound (non-offered) ≈ 7 –9; lower $<$ upper (bounds do not cross). Premium routes carry higher bounds. Fitted $F_j = \phi'W_j$ increasing in frequency, premium, and service; $\sim 86\%$ of inequalities satisfied.

2. Part 2 — interpretation (model answers)

- (a) *Fare coefficient and substitution.* The fare coefficient is negative and own-fare elasticities are around -2 to -3.5 , so demand is elastic. Plain logit imposes IIA: when one flight's fare rises, its lost travelers spread across all rivals in proportion to market share, regardless of how similar the flights are. That is wrong for a merger — it would predict the same substitution to a distant rival as to a near-identical one, mis-stating both the repricing incentive and which flights the merged firm would cancel. The random fare coefficient plus the income interaction breaks IIA and lets close flights be close substitutes.
- (b) *Diversion.* D_{jk} is the fraction of flight j 's lost travelers who switch to flight k . On a head-to-head route the AA-BB pair is the overlap, and $\bar{D}_{AB} = \frac{1}{2}(D_{jk} + D_{kj})$ measures how close the two flights are. It is not a selection device (every shared-route pair is an overlap); it is the magnitude that predicts recapture, and hence how much withdrawal pressure the route sees in Part 3. Larger $\bar{D}_{AB} \Rightarrow$ more traffic recaptured by cancelling one.
- (c) *Costs and Corts.* Marginal cost rises with fuel, distance, frequency, and premium; service is (correctly) ≈ 0 . The floored- mc share is 0%: no recovered markup exceeds its fare. A large floored share would mean the model implies markups above prices — economically impossible under the assumed Bertrand-Nash conduct — which rejects the demand/conduct specification (the Corts critique that conduct and demand are jointly tested here).
- (d) *Why offered $\rightarrow$ upper, non-offered $\rightarrow$ lower.* A carrier offers a route only if its expected variable profit covers the route's fixed cost, so for an *offered* route $F_j \leq \Delta VP_j^{\text{pre}}$ (upper bound). It stays out of a feasible route only if the entry profit would *not* cover the fixed cost, so for a *non-offered* route $F_k \geq \Delta VP_k$ (lower bound). Selection runs in opposite directions for the two groups, so the offered upper bounds sit above the non-offered lower bounds and the two sets do not cross.
- (e) *The fixed-cost function.* $F_j = \phi'W_j$ is increasing in frequency, premium, and service: routes flown more often, with a premium cabin, or with higher service cost more to keep on the schedule. Premium routes in particular show higher bounds and a positive premium coefficient — consistent with the marginal-cost side, where premium also raises cost.

3. Part 3 — merger with route withdrawal

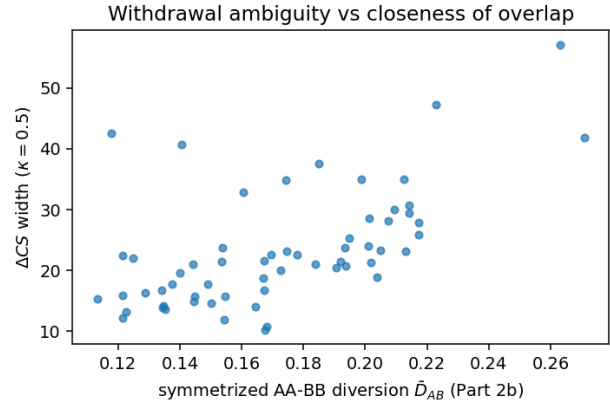
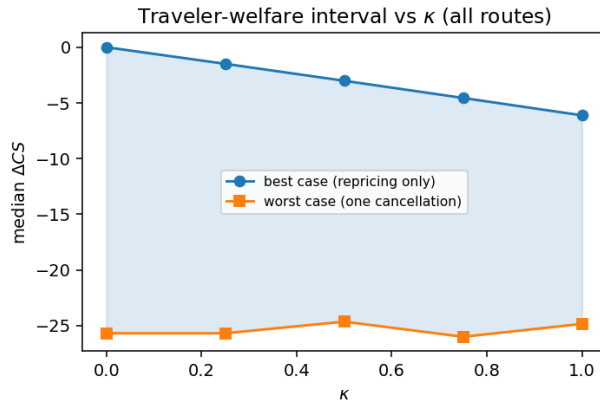
Mechanics (one route, R00_T0, $CS^{\text{pre}} = 75.6$). Both AA and BB flights are *at risk* at every κ : joint ownership lowers each flight's incremental value below its pre-merger fixed-cost ceiling. But route exit is not consistent — once one flight is cancelled its sibling recaptures the traffic and rises above its bound — so the consumer-worst *consistent* outcome is a single cancellation. Per-route interval (repricing = $CS^{\text{post}}(\mathcal{J}) - CS^{\text{pre}}$, withdrawal = $CS^{\text{post}}(\text{drop one}) - CS^{\text{post}}(\mathcal{J})$):

κ	repricing	withdrawal	ΔCS interval
0.00	0.00	−29.5	[−29.5, 0.00]
0.25	−2.10	−27.4	[−29.5, −2.10]
0.50	−4.24	−25.3	[−29.5, −4.24]
0.75	−6.41	−23.1	[−29.5, −6.41]
1.00	−8.59	−20.9	[−29.5, −8.59]

Full sweep (all 60 head-to-head routes $\times \kappa$; solution_part3.py). Medians across routes:

κ	med. repricing	med. withdrawal	med. width	med. worst ΔCS
0.00	0.00	−25.7	25.7	−25.7
0.25	−1.49	−23.9	23.9	−25.7
0.50	−3.01	−21.5	21.5	−24.6
0.75	−4.55	−20.9	20.9	−26.0
1.00	−6.11	−18.6	18.6	−24.8

Across routes the at-risk set is both flights on 253 of 300 (route, κ) cells, one flight on 43, and empty on 4. The correlation between the interval width and the Part 2(b) diversion $\bar{D}_{AB}$ (at $\kappa = 0.5$) is **+0.59**.



Expected interpretation. The point-identified *repricing* loss grows with κ (more fare coordination), while the set-identified worst case (cancel one flight) is roughly κ -invariant, so the lower endpoint stays near −25 and the interval *narrows* as κ rises. At $\kappa = 0$ repricing is exactly zero, so the *entire* potential harm is withdrawal: the portfolio-rationalization channel operates with no fare coordination at all. The positive width–diversion correlation is the payoff of Part 2(b): routes whose AA and BB flights are closer substitutes have larger recapture, hence larger and more ambiguous withdrawal harm. A strong submission gets these three points; the exact numbers may differ at the last decimal with N_DRAWS/seed, but the signs and the shape of both figures should match.